

One Curve Fitting: First-Order Transfer Function (Weighted)

Model

$$G(s) = \frac{b}{s+a} \Rightarrow G(j\omega) = \frac{b}{a+j\omega}$$

Data

$$H(\omega) = h(\omega)e^{j\phi(\omega)} = h(\omega)\cos\phi(\omega) + jh(\omega)\sin\phi(\omega)$$

Frequency Sweeping

$$H(\omega_k) = h_k e^{j\phi_k} = h_k c_k + j h_k s_k, \quad \text{where } k = 1, 2, \dots, n$$

where $c_k = \cos\phi_k$, $s_k = \sin\phi_k$.

Error Definition

$$\begin{aligned}\varepsilon_k &= (a + j\omega_k)(h_k c_k + j h_k s_k) - b \\ &= [a h_k c_k + j \omega_k h_k c_k + j a h_k s_k + j^2 \omega_k h_k s_k] - b \\ &= [a h_k c_k - \omega_k h_k s_k - b] + j [a h_k s_k + \omega_k h_k c_k]\end{aligned}$$

Real part:

$$R_{\varepsilon_k} = -h_k s_k \omega_k + h_k c_k a - b$$

Imaginary part:

$$I_{\varepsilon_k} = h_k c_k \omega_k + h_k s_k a$$

Partial Derivatives

$$\begin{aligned}\Rightarrow R_{\varepsilon_k} \frac{\partial R_{\varepsilon_k}}{\partial a} &= [-h_k s_k \omega_k + h_k c_k a - b](h_k c_k) \\ &= h_k^2 c_k^2 a - h_k c_k b - h_k^2 s_k c_k \omega_k\end{aligned}$$

$$\begin{aligned}\Rightarrow I_{\varepsilon_k} \frac{\partial I_{\varepsilon_k}}{\partial a} &= [h_k c_k \omega_k + h_k s_k a](h_k s_k) \\ &= h_k^2 s_k^2 a + h_k^2 c_k s_k \omega_k\end{aligned}$$

$$\begin{aligned}\Rightarrow R_{\varepsilon_k} \frac{\partial R_{\varepsilon_k}}{\partial b} &= [-h_k s_k \omega_k + h_k c_k a - b](-1) \\ &= h_k s_k \omega_k - h_k c_k a + b\end{aligned}$$

$$\Rightarrow I_{\varepsilon_k} \frac{\partial I_{\varepsilon_k}}{\partial b} = 0$$

Cost Function

$$J = \sum_{k=1}^n w(\omega_k) (R_{\varepsilon_k}^2 + I_{\varepsilon_k}^2)$$

Weighting Function

$$w(\omega) = \frac{1}{(1 + (\omega^2/\omega_c^2))^p}, \quad p = \frac{1}{2} \text{ or } 1$$

Optimization Conditions

$$\begin{aligned} \frac{\partial J}{\partial a} = 0 &\Rightarrow \sum_{k=1}^n w(\omega_k) \left[R_{\varepsilon_k} \frac{\partial R_{\varepsilon_k}}{\partial a} + I_{\varepsilon_k} \frac{\partial I_{\varepsilon_k}}{\partial a} \right] = 0 \\ &\Rightarrow \sum_{k=1}^n w(\omega_k) [h_k^2 c_k^2 a - h_k c_k b - h_k^2 s_k c_k \omega_k + h_k^2 s_k^2 a + h_k^2 c_k s_k \omega_k] = 0 \\ &\Rightarrow \sum_{k=1}^n w(\omega_k) [h_k^2 (c_k^2 + s_k^2) a - h_k c_k b] = 0 \\ &\Rightarrow \left(\sum_{k=1}^n w(\omega_k) h_k^2 \right) a - \left(\sum_{k=1}^n w(\omega_k) h_k c_k \right) b = 0 \end{aligned}$$

$$\begin{aligned} \frac{\partial J}{\partial b} = 0 &\Rightarrow \sum_{k=1}^n w(\omega_k) \left[R_{\varepsilon_k} \frac{\partial R_{\varepsilon_k}}{\partial b} + I_{\varepsilon_k} \frac{\partial I_{\varepsilon_k}}{\partial b} \right] = 0 \\ &\Rightarrow \sum_{k=1}^n w(\omega_k) [h_k s_k \omega_k - h_k c_k a + b] = 0 \\ &\Rightarrow - \left(\sum_{k=1}^n w(\omega_k) h_k c_k \right) a + \left(\sum_{k=1}^n w(\omega_k) \right) b = - \sum_{k=1}^n w(\omega_k) h_k s_k \omega_k \end{aligned}$$

Matrix Form

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

where:

$$a_{11} = \sum_{k=1}^n w(\omega_k) h_k^2$$

$$a_{12} = - \sum_{k=1}^n w(\omega_k) h_k c_k$$

$$a_{21} = a_{12}$$

$$a_{22} = \sum_{k=1}^n w(\omega_k)$$

$$y_1 = 0$$

$$y_2 = - \sum_{k=1}^n w(\omega_k) h_k s_k \omega_k$$

Solution

$$a = \frac{-a_{12}y_2}{a_{11}a_{22} - a_{12}^2}$$

$$b = \frac{a_{11}y_2}{a_{11}a_{22} - a_{12}^2}$$

Transfer Function

$$G(s) = \frac{b}{s + a}$$